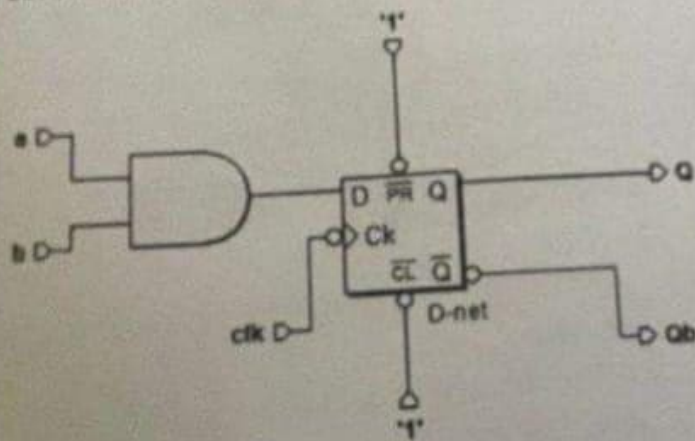


Q1. Write the corresponding VHDL code, for the Following circuit?



Q4. Correct the syntax error for the Following vhdl code?

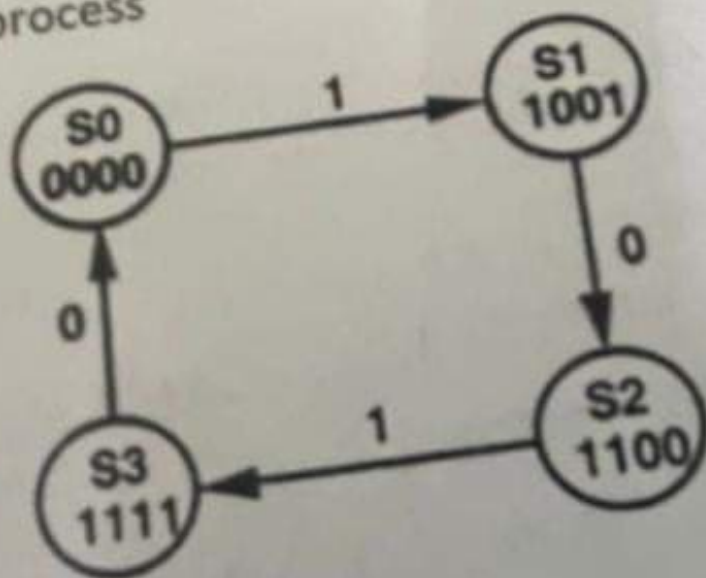
Q5- Draw the FSM for the current VHDL code?

Number of error	
-----------------	--

```
library IEEE;
use IEEE.std_logic_1164.all;

entity fsm is
  port ( clk, reset, x1 : IN std_logic;
        outp : OUT std_logic);
end entity;
architecture beh1 of fsm is
  type state_type is (s1,s2,s3,s4);
  signal state: state_type ;
begin
  process (clk,reset)
  begin
    if (reset ='1') then
      state <= s1; outp <= '1';
    elsif (clk='1' and clk'event) then
      case state is
        when s1 => if x1='1' then
state <= s2;
                    else      state <= s3;
                    end if;
                    outp <= '1';
        when s2 => state <= s4; outp
<= '1';
        when s3 => state <= s4; outp <=
'0';
        when s4 => state <= s1; outp <=
'0';
      end case;
    end if;
  end process;
end beh1;
```

Q3) write the vhdl code for the following FSM machine using the process



2- Draw the logic circuit for the current VHDL code?

```

LIBRARY ieee;
USE ieee.std_logic_1164.all;
USE ieee.numeric_std.all;
ENTITY aaa IS
  PORT(
    --> Inputs:
    S1: IN std_logic;
    S2: IN std_logic;
    S3: IN std_logic;
    --> Outputs:
    d0,d1,d2,d3,d4,d5,d6,d7: OUT
    std_logic );
END aaa;
ARCHITECTURE structural OF aaa IS
  Components:
  COMPONENT NOT_gate IS
    PORT( I: IN std_logic;
          O: OUT std_logic );
  END COMPONENT;
  --
  COMPONENT AND3_gate IS
    PORT( I0,I1,I2: IN std_logic;
          O: OUT std_logic );
  END COMPONENT;
  --> Signals:
  SIGNAL S001: std_logic;
  SIGNAL S002: std_logic;
  SIGNAL S003: std_logic;
  BEGIN -- structural
  -->
  Component Mapping:
  C073: NOT_gate PORT MAP ( S1, S001 );
  C072: NOT_gate PORT MAP ( S2, S002 );
  C074: NOT_gate PORT MAP ( S3, S003 );

  c141: AND3_gate PORT MAP ( S001, S002,
    S003,d0 );
  C142: AND3_gate PORT MAP ( S001, S002,
    S3,d1 );
  C143: AND3_gate PORT MAP ( S001, S2,
    S003,d2 );
  C144: AND3_gate PORT MAP ( S001, S2,
    S3,d3 );
  C145: AND3_gate PORT MAP ( S1, S002,
    S003,d4 );
  C146: AND3_gate PORT MAP ( S1, S002,
    S3,d5 );
  C147: AND3_gate PORT MAP ( S1, S2,
    S003,d6 );
  C148: AND3_gate PORT MAP ( S1, S2,
    S3,d7 );
END structural;
library ieee;
use ieee.std_logic_1164.all;

ENTITY AND3_gate IS
  PORT( I0,I1,I2: IN std_logic;
        O: OUT std_logic );
END AND3_gate;

ARCHITECTURE behavioral OF AND3_gate IS
  BEGIN
    O <= (I0 and I1 and I2 );
  END behavioral;
library ieee;
use ieee.std_logic_1164.all;

ENTITY NOT_gate IS
  PORT( I: IN std_logic;
        O: OUT std_logic );
END NOT_gate;

ARCHITECTURE behavioral OF NOT_gate IS
  BEGIN
    O <= (not I);
  END behavioral;
  
```